

Laboratory 4 (100 Points)

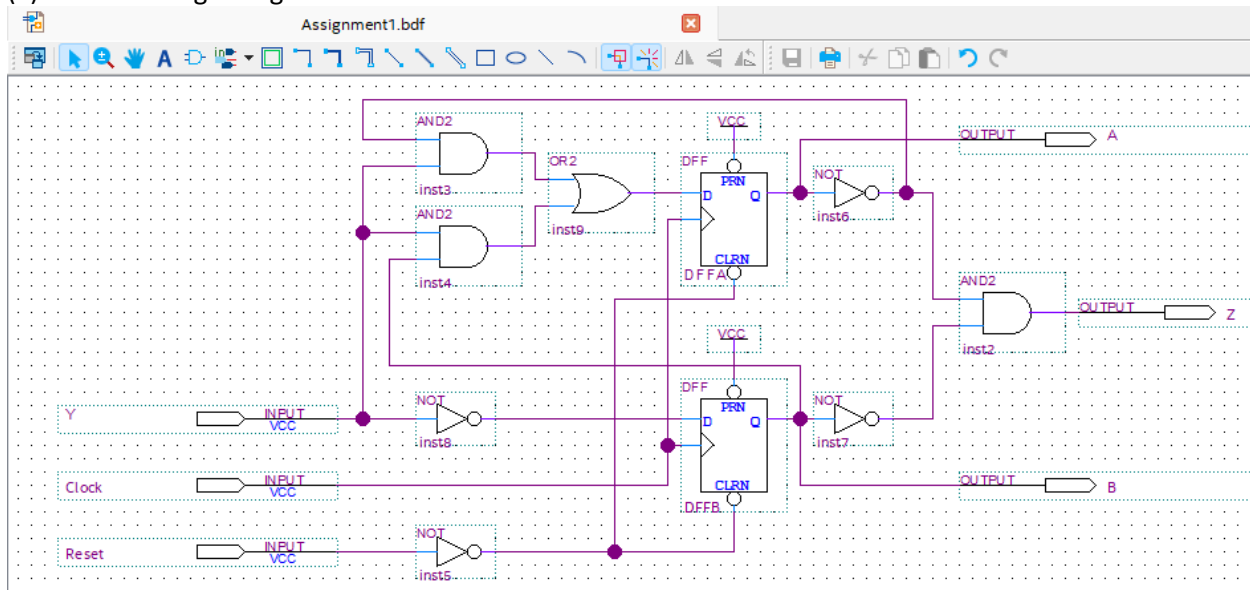
Assignment 1 (20 Points)

(4-5) A sequential circuit with two **D flip-flops** A and B, one input Y, and one output Z is specified by the following input equations:

$$D_A = BY + \bar{A}Y, \quad D_B = \bar{Y}, \quad Z = \bar{A}\bar{B}$$

- Draw the logic diagram of the circuit.
- Derive the state table.
- Derive the state diagram.
- Simulate the circuit:
 - Initial state: 00
 - Input sequence: 0100
 - Check the output Z with the state table

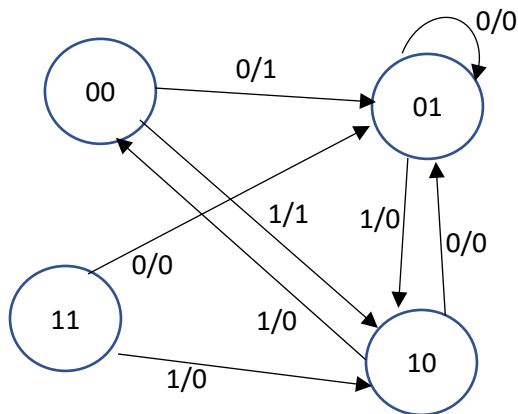
(a) Draw the logic diagram of the circuit



(b) Derive the state table.

Present State		Input	Next State		Output
A	B	Y	A	B	Z
0	0	0	0	1	1
0	0	1	1	0	1
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	0	0
1	1	0	0	1	0
1	1	1	1	0	0

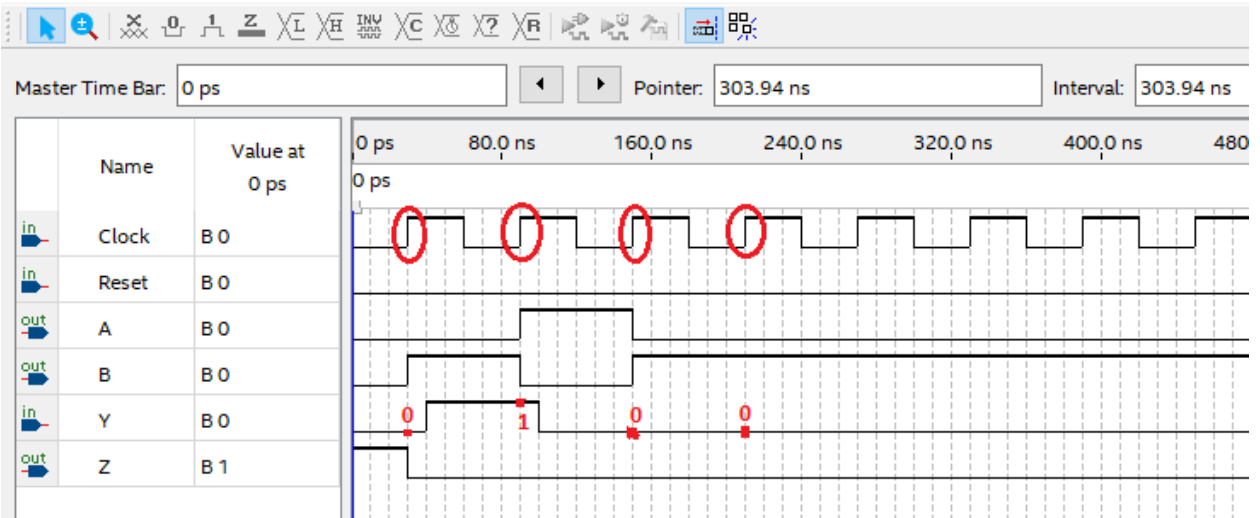
(c) Derive the state diagram.



(d) Simulate the circuit with initial state AB = 00 and input sequence Y = 0100

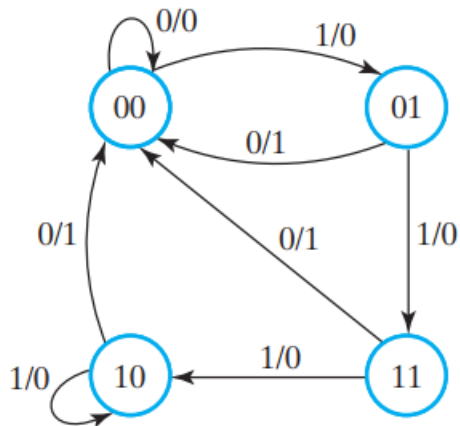
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Assignment 2 (20 Points)

The state diagram of a sequential circuit is shown in the following figure. Design the circuit with D flip-flops using binary code state assignment. Simulate the circuit.



- (a) Derive the state table from the above state diagram
- (b) Draw the logic diagram of the circuit.
- (c) Simulate the circuit and check the output with initial state AB = 00 and input sequence X = 0100

Assignment 3 (20 Points)

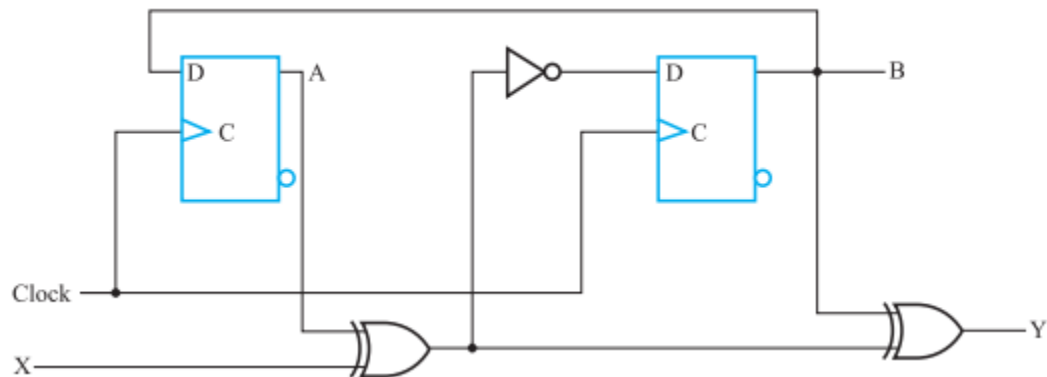
(4-6) A sequential circuit with two D flip-flops A and B, two inputs X and Y, by the following input equations:

$$D_A = XA + \overline{X}Y, \quad D_B = XB + XA$$

- (a) Draw the logic diagram of the circuit.
- (b) Derive the state table.
- (c) Derive the state diagram.
- (d) Simulate the circuit:
 - Initial state: 00
 - Input sequence: X = 00110011
Y = 01010101

Assignment 4 (10 Points)

(4-11) A sequential circuit has two D flip-flops, one input X, and one output Y. The logic diagram of the circuit is shown in the following figure. Derive the state table and state diagram of the circuit.



Assignment 5 (10 Points)

Design a 2-bit down counter. The 2-bit down counter is a sequential circuit with three D flip-flops A, B and C, and one input X. When X = 0, the state of the circuit is reset to 00. When X = 1, the circuit goes through the state transitions from 11 to 10 to 01 to 00, back to 11, and then repeats.

- Derive the state diagram
- Derive the state table
- Derive the necessary equations
- Draw the logic diagram of the circuit
- Simulate the circuit with the following requirements
 - Initial state: 11
 - Input sequence X: 101101

Assignment 6 (20 Points)

Design a sequence detector for the bit pattern "011" which has one serial input X, and one output Y.

The detector asserts $Y = 1$ when the previous input value "011" is recognized; overlapping occurrences are allowed. Assume that the rising-edge-triggered clock is used.

- Derive the state diagram.
- Derive the state table.
- Derive simplified expressions.
- Design the logic diagram.
- Simulate the circuit with the following requirements.
 - Initial state: 0 for all flip-flops
 - Input X: 011011